

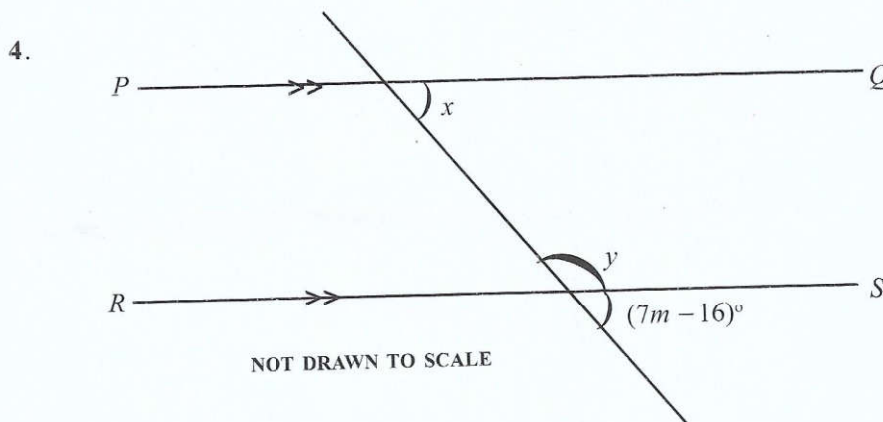
SECTION A
[40 marks]

Answer **all** the questions in this section. All questions carry **equal** marks.

1. The third and eighth terms of a Geometric Progression (G.P) are 4 and $\frac{1}{256}$ respectively. Find the:
 - (a) common ratio;
 - (b) first term,
 - (c) sum of the **first** and **second** terms.

2. In a school election for the position of SRC President, one candidate obtained 87.5 % of the votes cast. The other candidates obtained a total of 275 votes.
 - (a) How many votes did the winner obtain.
 - (b) If the voter turnout was 55 % , find the total number of eligible voters in the school.

3. A mason hired a labourer to fill a cubic rubber tank of length 1.2 m with water. The labourer used a container of volume 54 litres. If the mason agreed to pay GH¢ 1.75 for a container of water fetched, calculate:
 - (a) in litres, the volume of the tank;
 - (b) the number of containers of water that can fill the tank;
 - (c) the amount of money paid to the labourer to fill the tank with water.



In the diagram $\overline{PQ} \parallel \overline{RS}$. Given that $x : y = 2 : 7$, find the value of m .

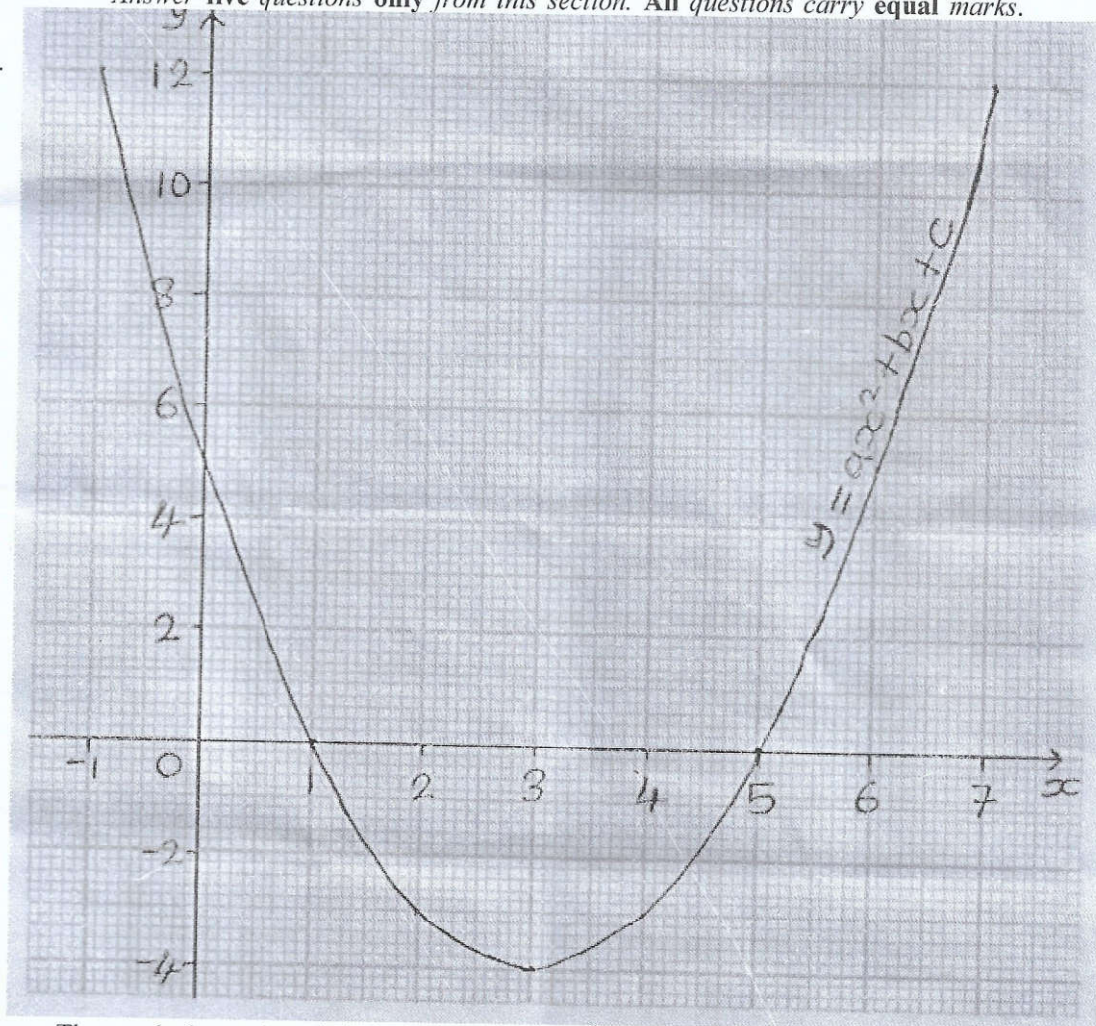
5. (a) Given that $\sin y = \frac{5}{13}$, $0^\circ \leq y \leq 90^\circ$, evaluate $\frac{3\cos y - 2\sin y}{6\tan y}$.
- (b) A number is chosen at random from the set of integers, 10 to 30 inclusive. Find the probability that the number is a multiple of 3 or 4.

SECTION B

[60 marks]

Answer **five** questions **only** from this section. All questions carry equal marks.

6.



The graph shows the relation of the form $y = ax^2 + bx + c$, where a , b and c are constants.

- (a) State the scale used for **each** axes.
- (b) Find the values of a , b and c .
- (c) Find the values of x when $y = 7$.
- (d) Write the coordinates of the minimum point.
- (e) State the range of values of x for which $y < 0$.

7. The estimated cost of building a two bedroom house is GH¢ 120,000.00. It is made up of labour, materials and consultancy fee in the ratio 5: 12 : 3 respectively. At the start of the construction, the labour cost increased by $2y\%$, cost of materials increased by $2.5y\%$ while consultancy fee remain unchanged. If the new cost of labour is two-fifth the new cost of materials, find the:

- (a) value of y ;
(b) new cost of building the house.

8. (a) Kojo can do a piece of work in 5 hours. Edna can do the same piece of work in 4 hours. How long will it take the two of them to do the same work together, if they worked at the same rate?

- (b) A solid right triangular prism of length 12 cm and a cross-section which is an equilateral triangle of side 6 cm. Find, correct to **four** significant figures, the total surface area.

9. (a) Copy and complete the table of values for $y = \cos x - 3 \sin x$, $0^\circ \leq x \leq 180^\circ$.

x	0°	20°	40°	60°	80°	100°	120°	140°	160°	180°
y	1.0			-2.1					-2.0	

- (b) Using a scale of 2 cm to 20° on the x axis and 4 cm to 1 unit on the y axis, draw the graph of $y = \cos x - 3 \sin x$, $0^\circ \leq x \leq 180^\circ$.

- (c) Using the graph, find the:

- (i) truth set of $2 + \cos x = 3 \sin x$;
(ii) range of values of x for which y increases as x increases;
(ii) minimum point of the curve.

10.

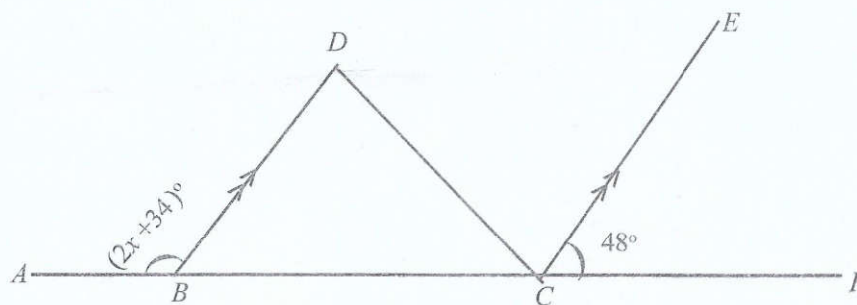
Age (years)	7	8	9	10	11	12
Number of children	$2x$	$3x$	$4x - 1$	x	$x - 2$	$x - 3$

The table shows the ages in years of 42 children at a birthday party.

- (a) Find the value of x .
(b) Calculate, correct to the **nearest** whole number, the mean age.
(c) Find the probability of selecting at random a child whose age is **not** less than 9 years.

11. (a) Given that $\cos(2x + 15^\circ) = \sin(x - 30^\circ)$, $0^\circ \leq x \leq 90^\circ$, find the value of x .

(b)



NOT DRAWN TO SCALE

In the diagram, $ABCF$ is a straight line, $\overline{BD} \parallel \overline{CE}$, $\angle ABD = (2x + 34)^\circ$. If $\angle BCD = \frac{1}{3} \angle BDC$ and $\angle FCE = 48^\circ$, find:

- (i) the value of x ;
 - (ii) $\angle BDC$.
12. (a) Fifteen persons were shortlisted for a job interview. On the day of the interview, each applicant was assigned a number from 1 to 15. An applicant was called at random from the list of numbers. Find the probability that the applicant called has the number which is:
- (i) prime;
 - (ii) multiple of 3;
 - (iii) divisible by 5.
- (b) $PQRS$ is a quadrilateral with $P(4, 4)$, $S(8, 8)$ and $R(12, 8)$. If $\overrightarrow{PQ} = 4\overrightarrow{SR}$, find the coordinates of Q .
13. (a) The hour hand of a 24 hour clock moves from 13:15 p. m. to 17:45 p. m. If the length of the hour hand is 3.5 cm, find correct to **three** significant figures, the area swept by the hour hand. [Take $\pi = \frac{22}{7}$]
- (b) The coordinates of two points M and N in a plane are $(-4, 3)$ and $(x, 6)$ respectively. If $|\overline{MN}| = 3\sqrt{10}$ units, find the values of x .

END OF PAPER